Alice Sunghyun Park

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Mechanical and Aerospace Engineer

PROFESSIONAL SUMMARY

Mechanical Engineering student at Cornell ('28) with research experience spanning high-speed fluid-dynamics experiments and physics-based computation. Conducted high-speed imaging studies at Seoul National University and developed post-processing pipelines to extract quantitative flow metrics; currently applying N-body modeling and data visualization to electrospray plume simulations and designing a scaled linear induction motor for Cornell Hyperloop's mini-pod. Interested in aerodynamics, propulsion, and computational mechanics roles in aerospace and performance-driven systems.

EDUCATION

Cornell University, College of Engineering — Ithaca, NY
B.E. Mechanical Engineering & Computer Science minor • GPA: 3.65/4.00 • Class of 2028
Relevant Coursework: Object-Oriented Programming & Data Structures (Java), Functional Programming (OCaml), Mechanical Design, Dynamics, Thermodynamics, Statics & Mechanics of Solids, Electricity & Magnetism, Mechanics, Linear Algebra, Differential Equations, Multivariable Calculus, etc.

Berlin Metropolitan School — Berlin, Germany
High School Diploma • International Baccalaureate Diploma Programme: 41/45

TECHNICAL SKILLS

Programming & Data: Python, MATLAB, Java, C/C++, OCaml; Jupyter Notebooks; image processing, numerical analysis, and simulation scripting
CAD & Simulation: SolidWorks, Fusion, Onshape; ANSYS (Mechanical, Maxwell) — meshing, boundary-condition setup, electromagnetic and structural simulation
Lab, Test & Fabrication: High-speed imaging; event-based vision (Prophesee DVS); Arduino; mill, lathe, bandsaw; 3D printing, laser cutting
Documentation: Git/GitHub; Microsoft Excel, PowerPoint, Word; LaTeX
Languages: English (fluent), Korean (fluent), German (intermediate), French (beginner)

RESEARCH EXPERIENCE

Multiphase Flow and Flow Visualization Lab (MFFV), *Research Intern at Seoul National University* — *Seoul, South Korea • June 2025 – August 2025, December 2025 – January 2026*

- Conducted high-speed imaging (≈2,000 fps) of multiphase and granular-wetting experiments to capture wetting-front propagation, void dynamics, and saturation patterns in polymer granular beds; developed custom post-processing pipelines in Python, MATLAB, and C/C++ to extract quantitative flow metrics from image sequences.
- Designed reproducible image-processing and analysis pipelines (background subtraction, denoising, edge/contour detection, wetting-front tracking) to convert high-speed imaging data into time-resolved flow metrics for cross-condition comparison.
- Co-author, APS DFD 2025 (Houston, TX): contributed experimental visualization and computational post-processing for *Experimental visualization of sticking & drying in granular beds* (accepted, Sep 2025).

ASTRA Lab, ***Undergraduate Researcher at Cornell University*** — *Ithaca, NY • August 2025 – Present*

- Post-processed and validated core components of an N-body solver using analytic checks and timestep/grid-sensitivity tests to confirm numerical stability.
- Developed a Python-based visualization toolkit to read VTP outputs and generate publication-quality figures, including particle trajectories, species-density histograms (neutrals through trimers), and time-of-flight curves.
- Executed and managed C++ electrospray simulations to generate datasets for parametric and visual analysis supporting ongoing investigation.
- Designed and executed convergence studies by modifying and running C++ electrospray simulations within the lab's N-body code, performing timestep and mesh-resolution sweeps to assess temporal accuracy, numerical stability, and solution convergence.
- Co-author, AIAA SciTech 2026**:** *Kinetic Plume Modeling of Electrospray Ion Sources with Emitter and Extractor Defects* — contributed simulation setup, convergence validation, parametric analysis, and figure generation.
- Contributing to an ongoing journal manuscript on electrospray N-body plume simulations by conducting parametric studies and synthesizing results into figures and written analysis, with emphasis on particle-species distributions, plume divergence, and sensitivity to emitter offset and fabrication-induced asymmetries.

ENGINEERING PROJECTS

Cornell Hyperloop, *Magnetic Subteam Member* — *Ithaca, NY • September 2024 – Present*

- Engineered a magnetic-force test assembly (I-beam frame, translating stage, distance sensing) to evaluate off-the-shelf transformer performance; modeled components in SolidWorks and simulated side-rail force loads using ANSYS Mechanical.
- Leading development of a scaled single-sided linear induction motor (LIM) for pod propulsion, researching induction principles and design constraints, modeling geometries in SolidWorks, and simulating magnetic field interactions with the secondary in ANSYS Maxwell.

ADDITIONAL PROJECTS

European Aviation Carbon Emissions (EUA) – *Research for IBDP Extended Essay* — *Berlin, Germany • August 2022 – February 2023*

- Evaluated the effectiveness of the EU Aviation Emissions Trading System (ETS) in reducing CO_2 emissions using EEX and EEA datasets, producing evidence-based visual analyses.
- Integrated EUA auction price and volume data with aviation-sector emissions to perform sensitivity and trend analyses and generate reproducible assessment figures.

"Every Sound is Precious" (어느 소리 하나 귀하지 아니한 것 없다**) – *Author, Ddi*** — *Berlin, Germany • February – October 2023*

- Authored and published a Korean-language poetry collection integrating classical music storytelling; managed drafting, editing, and layout through final publication (Oct. 12, 2023).

LEADERSHIP & ACTIVITIES

Society of Women Engineers (SWE) — Committee Member, Alumni & Faculty Relations — *Ithaca, NY • September 2025 – Present*

- Organize large-scale student–faculty networking events, including the Spring 2025 student–faculty dinner, and coordinate the Alumni Mentorship Program to connect students with industry and academic mentors.